



CONDITION BASED ASSESSMENT

FULL SCANNING REPORT

SBUM DIAGNOSTIC PAHANG

TNB BAHAGIAN PEMBAHAGIAN PAHANG

FUNCTIONAL LOCATION (ERMS)	:	CKTN/PCE/J01080
FUNCTIONAL LOCATION (SITE)	:	CKTN/PCE/J01080
SUBSTATION NAME (ERMS)	:	KK3 PAYA BESAR
SUBSTATION NAME (SITE)	:	KK3 PAYA BESAR
COORDINATE	:	3.788662, 103.239089
SUBSTATION TYPE	:	PCE
AREA	:	KUANTAN
STATE	:	PAHANG
PO	:	42225439
DATE	:	03 MAR 2026

SCANNED BY	
THERMOGRAPHY	: MUHAMMAD ADAM BIN SAFARUDDIN (2017MY34N004)

EQUIPMENT/TOOLS	
THERMOGRAPHY	: FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY	
NAME OF CONTRACTOR	: EET ENGINEERING SERVICES SDN BHD
ADDRESS	: NO.38, JALAN SS19/1D,
	: 47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



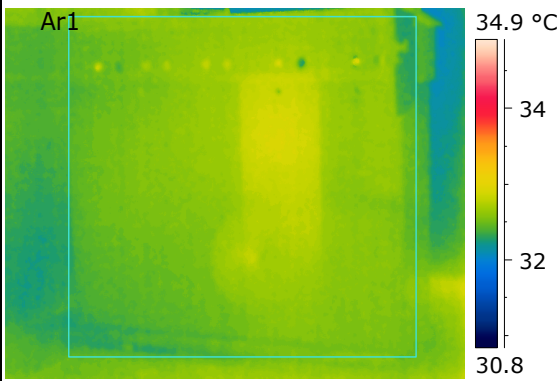
EXECUTIVE SUMMARY

NO.	EQUIPMENT	DEFECT AREA	IR (Abs T/ Δ T)	DEFECT
1.	RMU SF6, INDKOM	OVERVIEW	-	-
		OVERVIEW TOP	-	-
	PANEL CKN03071 KRAFTANGAN	CABLE COMPARTMENT	-	
	PANEL CKN03074 TMN. PANDAN INDAH	CABLE COMPARTMENT	-	
	PANEL CKN03779 RESTORAN MARRYBROWN	CABLE COMPARTMENT	-	
	PANEL CKN03072 TX1 1000KVA	FUSE COMPARTMENT	-	
	PANEL CKN03073 TX2 750KVA	FUSE COMPARTMENT	-	
2.	TX1 - SGB 1000KVA	OVERVIEW	-	-
		OVERVIEW TOP	-	-
		HV BUSHING	-	
		HV CABLES	-	
		LV BUSHING	-	
		LV CABLES	-	
3.	TX2 – EWT 750KVA	OVERVIEW	-	-
		OVERVIEW TOP	-	-
		HV BUSHING	-	
		HV CABLES	-	
		LV BUSHING	-	
		LV CABLES	-	
4.	FP TX1	OVERVIEW	-	-
5.	FP TX2	OVERVIEW	-	-

VISUAL DEFECT SUMMARY

NO.	EQUIPMENT	DEFECT DESCRIPTION	ADDITIONAL REMARKS
1	SUBSTATION	CPR POSTER OLD VERSION	SWG ROOM
2	FP/LVDB	NO FEEDER NO./FEEDER NAME	FP TX1 - NOT FOLLOW NEW STANDARD
3	FP/LVDB	NO FEEDER NO./FEEDER NAME	FP TX2 - NOT FOLLOW NEW STANDARD

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	OVERVIEW			Panel No. -
Panel Name		-			

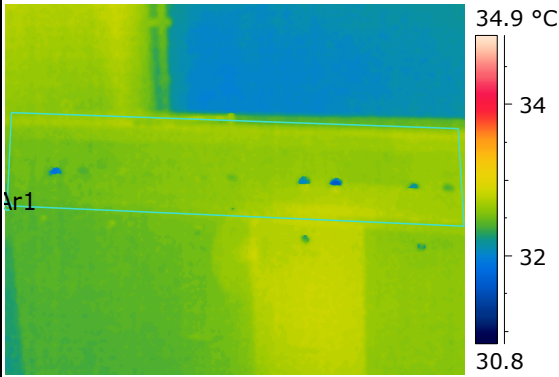
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	32.6 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	-	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	OVERVIEW TOP			Panel No.
Panel Name	-				

IR IMAGE	VISUAL IMAGE																												
																													
<table> <tr> <th>PARAMETER</th><th>VALUE</th></tr> <tr> <td>Spot Temp.1 (Sp1)</td><td>-</td></tr> <tr> <td>Spot Temp. 2 (Sp2)</td><td>-</td></tr> <tr> <td>Average Temp. (Ar1)</td><td>32.6 °C</td></tr> <tr> <td>Ambient Temp.</td><td>33.1 °C</td></tr> <tr> <td>Delta T (Δ T)</td><td>-</td></tr> <tr> <td>Load (A)</td><td>-</td></tr> <tr> <td>Defect</td><td>-</td></tr> </table>	PARAMETER	VALUE	Spot Temp.1 (Sp1)	-	Spot Temp. 2 (Sp2)	-	Average Temp. (Ar1)	32.6 °C	Ambient Temp.	33.1 °C	Delta T (Δ T)	-	Load (A)	-	Defect	-	<table> <tr> <th>PARAMETER</th><th>VALUE</th></tr> <tr> <td>Serial No.</td><td>14093213346DE</td></tr> <tr> <td>Heater (A)</td><td>-</td></tr> <tr> <td>CB Position</td><td>-</td></tr> <tr> <td>Busbar Position</td><td>-</td></tr> <tr> <td>Cable Type</td><td>-</td></tr> </table>	PARAMETER	VALUE	Serial No.	14093213346DE	Heater (A)	-	CB Position	-	Busbar Position	-	Cable Type	-
PARAMETER	VALUE																												
Spot Temp.1 (Sp1)	-																												
Spot Temp. 2 (Sp2)	-																												
Average Temp. (Ar1)	32.6 °C																												
Ambient Temp.	33.1 °C																												
Delta T (Δ T)	-																												
Load (A)	-																												
Defect	-																												
PARAMETER	VALUE																												
Serial No.	14093213346DE																												
Heater (A)	-																												
CB Position	-																												
Busbar Position	-																												
Cable Type	-																												

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	CABLE COMPARTMENT		Panel No.	CKN03071
Panel Name		KRAFTANGAN			

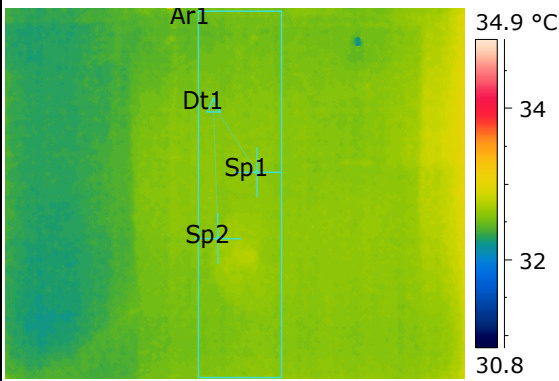
IR IMAGE		VISUAL IMAGE	
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	32.6 °C	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	32.5 °C	Heater (A)	-
Average Temp. (Ar1)	32.6 °C	CB Position	CLOSE
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.1 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	CABLE COMPARTMENT		Panel No.	CKN03074
Panel Name		TMN. PANDAN INDAH			

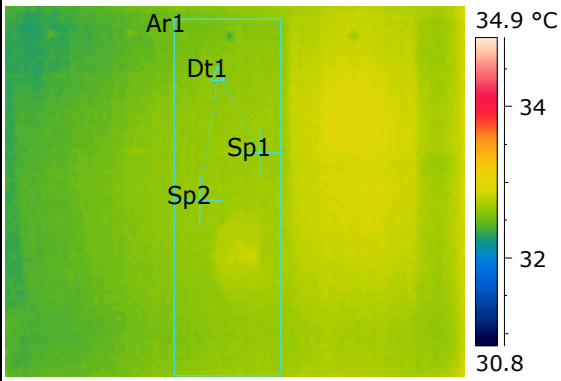
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	32.6 °C	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	32.7 °C	Heater (A)	-
Average Temp. (Ar1)	32.6 °C	CB Position	CLOSE
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	0.0 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	CABLE COMPARTMENT		Panel No.	CKN03779
Panel Name		RESTORAN MARRYBROWN			

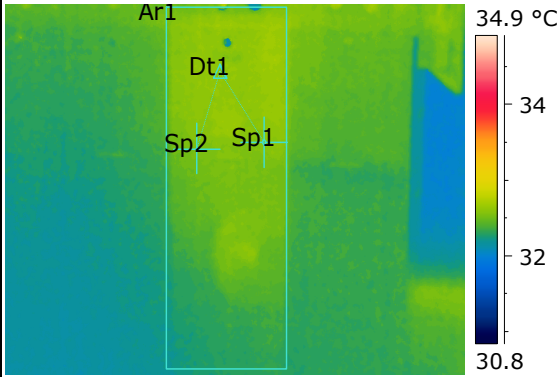
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	32.7 °C	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	32.7 °C	Heater (A)	-
Average Temp. (Ar1)	32.6 °C	CB Position	CLOSE
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	0.1 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	FUSE COMPARTMENT		Panel No.	CKN03072
Panel Name		TX1 1000KVA			

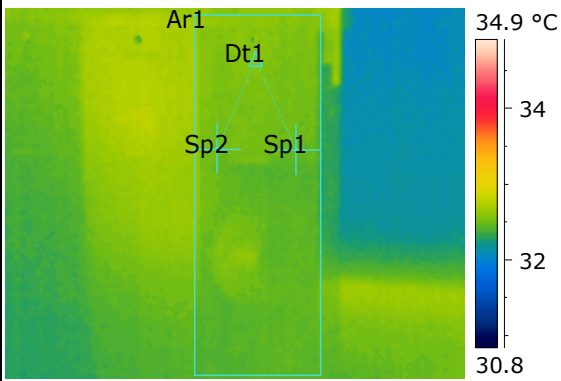
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	32.6 °C	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	32.5 °C	Heater (A)	-
Average Temp. (Ar1)	32.5 °C	CB Position	CLOSE
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	0.2 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	RMU SF6	Manufacturer	INDKOM	Time: 11:03 AM
	Model	INS24	Rating	12kV	Humidity: 70%
	Area	FUSE COMPARTMENT		Panel No.	CKN03073
Panel Name	TX2 750KVA				

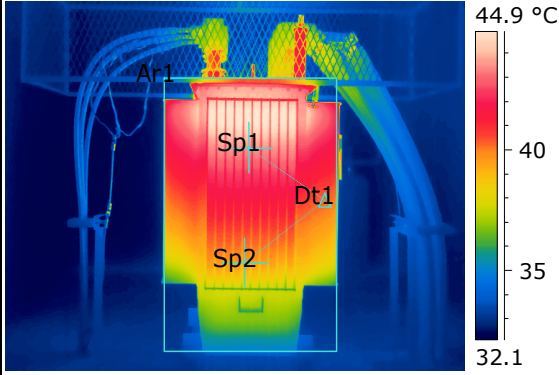
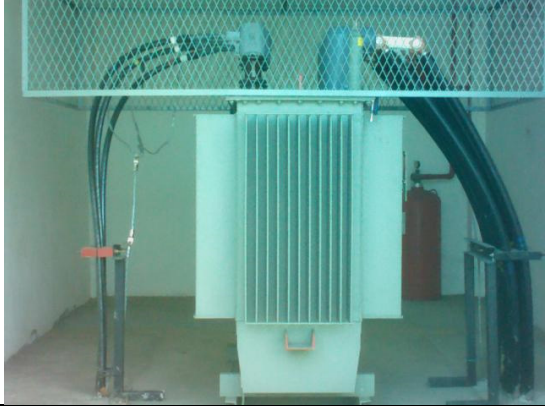
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	32.5 °C	Serial No.	14093213346DE
Spot Temp. 2 (Sp2)	32.5 °C	Heater (A)	-
Average Temp. (Ar1)	32.5 °C	CB Position	CLOSE
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.1 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	OVERVIEW		Tx No.	TX1
Location	-				

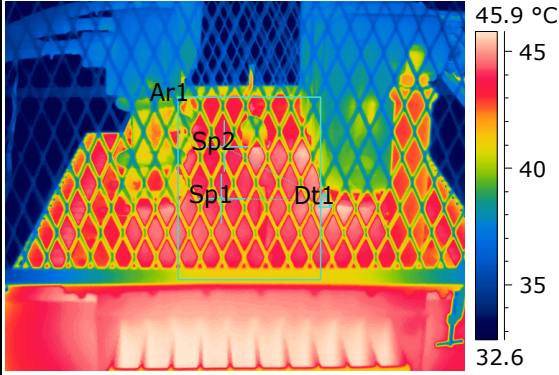
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	43.5 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	39.5 °C	Heater (A)	-
Average Temp. (Ar1)	39.5 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	4.0 °C	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	OVERVIEW TOP		Tx No.	TX1
Location		-			

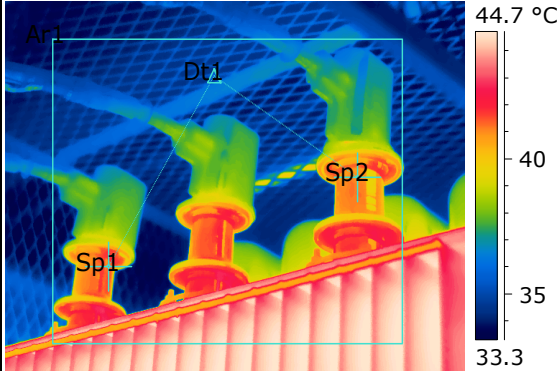
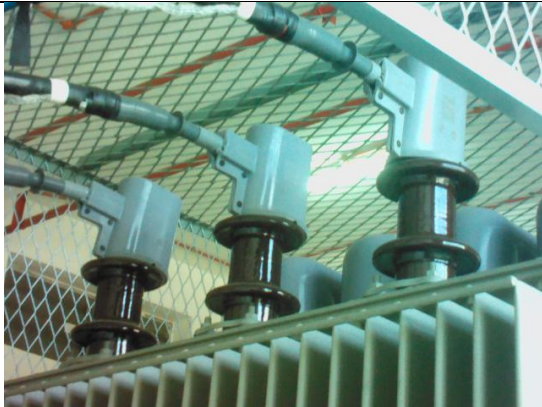
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	43.3 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	42.4 °C	Heater (A)	-
Average Temp. (Ar1)	42.4 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.9 °C	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

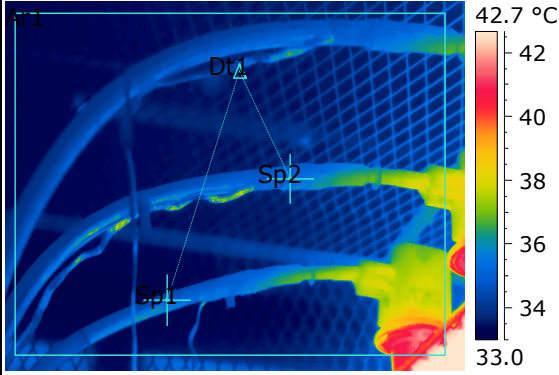
Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	BUSHING		Tx No.	TX1
Location	HV SIDE				

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	41.3 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	41.0 °C	Heater (A)	-
Average Temp. (Ar1)	38.1 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.3 °C	Cable Type	XLPE
Load (A)	-		
Defect			
Analysis & Recommendations:			
Analysis: No Anomaly.			
Recommendation: -			

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	CABLE		Tx No.	TX1
Location	HV SIDE				

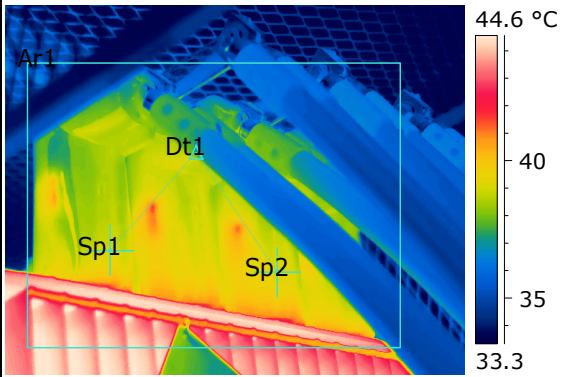
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	35.2 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	35.2 °C	Heater (A)	-
Average Temp. (Ar1)	34.3 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.0 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

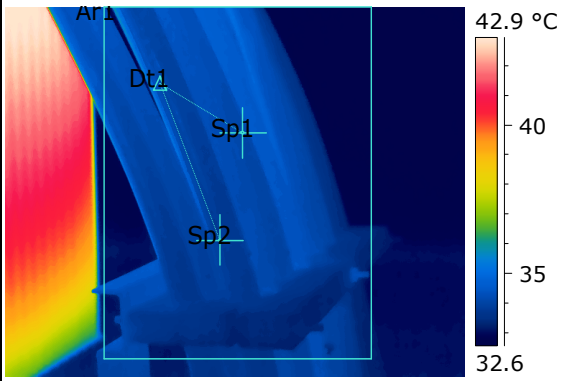
Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	BUSHING		Tx No.	TX1
Location	LV SIDE				

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	39.1 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	39.1 °C	Heater (A)	-
Average Temp. (Ar1)	38.1 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.0 °C	Cable Type	XLPE
Load (A)	-		
Defect			
Analysis & Recommendations:			
Analysis: No Anomaly.			
Recommendation: -			

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	SGB	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	1000KVA	Humidity: 70%
	Area	CABLE		Tx No.	TX1
Location	LV SIDE				

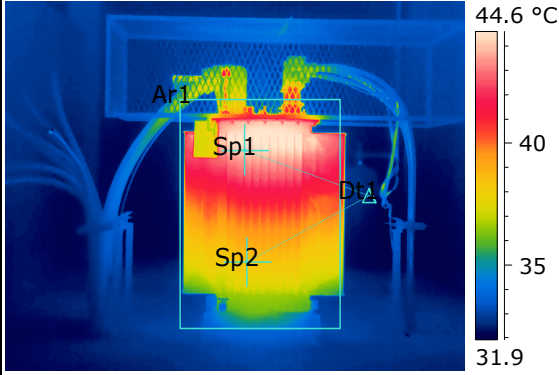
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	34.1 °C	Serial No.	718834
Spot Temp. 2 (Sp2)	33.9 °C	Heater (A)	-
Average Temp. (Ar1)	33.8 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.2 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	OVERVIEW		Tx No.	TX2
Location		-			

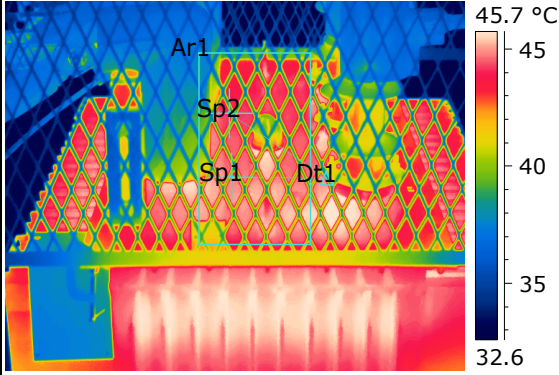
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	44.0 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	38.4 °C	Heater (A)	-
Average Temp. (Ar1)	38.8 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	5.6 °C	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	OVERVIEW TOP		Tx No.	TX2
Location		-			

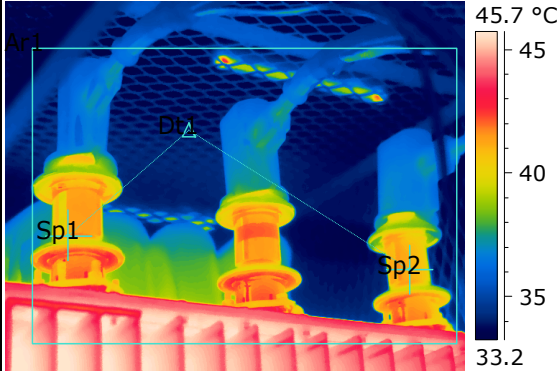
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	44.4 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	43.8 °C	Heater (A)	-
Average Temp. (Ar1)	42.2 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.7 °C	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

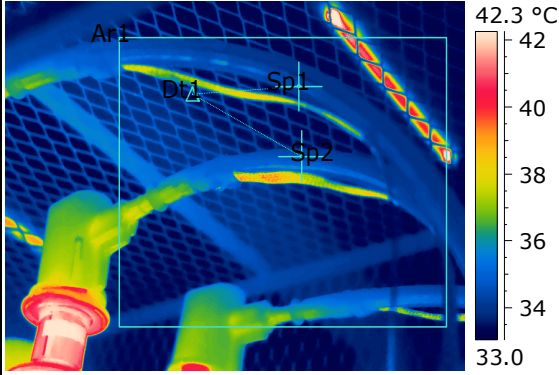
Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	BUSHING		Tx No.	TX2
Location	HV SIDE				

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	41.4 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	41.1 °C	Heater (A)	-
Average Temp. (Ar1)	37.4 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.3 °C	Cable Type	XLPE
Load (A)	-		
Defect			
Analysis & Recommendations:			
Analysis: No Anomaly.			
Recommendation: -			

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	CABLE			Tx No.
Location	HV SIDE				

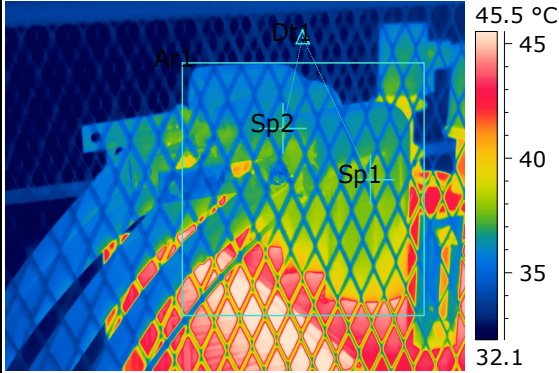
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	34.6 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	34.5 °C	Heater (A)	-
Average Temp. (Ar1)	34.4 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (Δ T)	0.1 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	BUSHING		Tx No.	TX2
Location	LV SIDE				

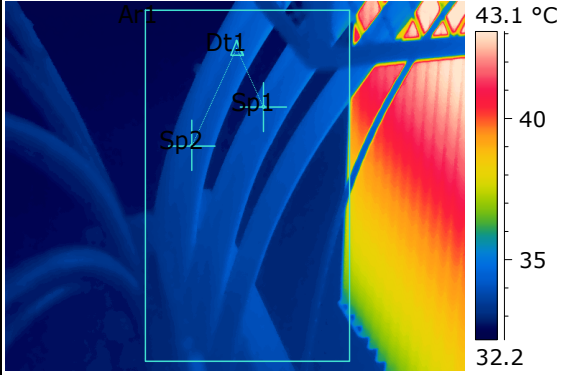
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	37.5 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	36.9 °C	Heater (A)	-
Average Temp. (Ar1)	37.7 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.6 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 11:03 AM
	Model	HERMETICALLY SEAL	Rating	750KVA	Humidity: 70%
	Area	CABLE			Tx No.
Location	LV SIDE				

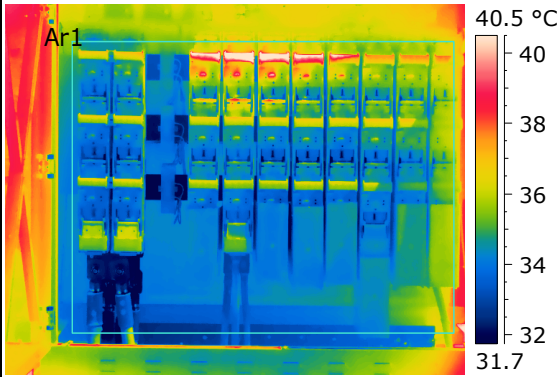
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	34.1 °C	Serial No.	EWT67313
Spot Temp. 2 (Sp2)	33.8 °C	Heater (A)	-
Average Temp. (Ar1)	33.5 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	0.3 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	FP TX1	Manufacturer	TOPRANK	Time: 11:03 AM
	Model	DIN TYPE	Rating	1600A	Humidity: 70%
	Area	OVERVIEW			Feeder No. -
Feeder Name		-			

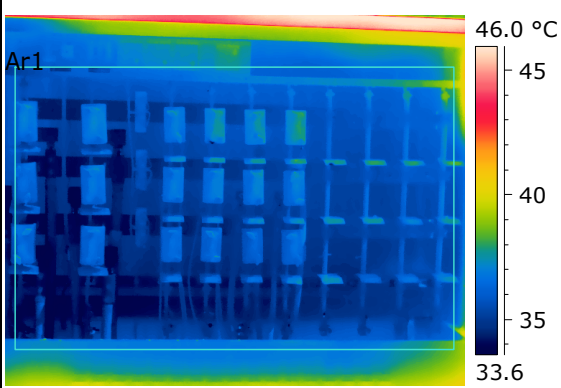
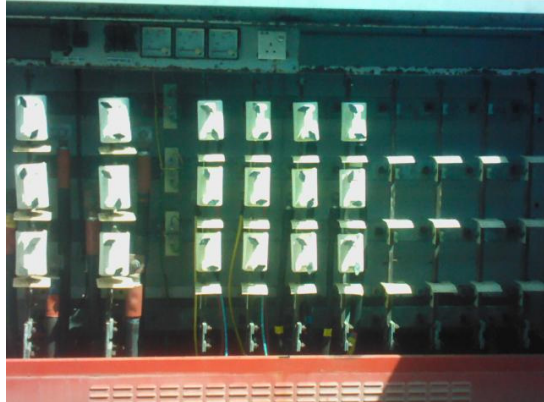
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	DFPAL89
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	34.5 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	KK3 PAYA BESAR			Date: 03/03/2026
	Equipment	FP TX2	Manufacturer	SSE	Time: 11:03 AM
	Model	J-SLOTTED	Rating	1600A	Humidity: 70%
	Area	OVERVIEW			Feeder No. -
Feeder Name		-			

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	FP9705-0278
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	35.6 °C	CB Position	-
Ambient Temp.	33.1 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

SUBSTATION CONDITION

SUBSTATION OVERVIEW



SIGNBOARD



SWITCHGEAR



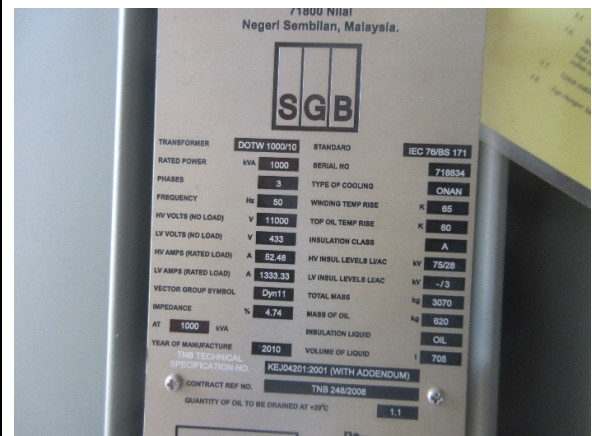
SWITCHGEAR NAMEPLATE



TRANSFORMER 1



TRANSFORMER 1 NAMEPLATE

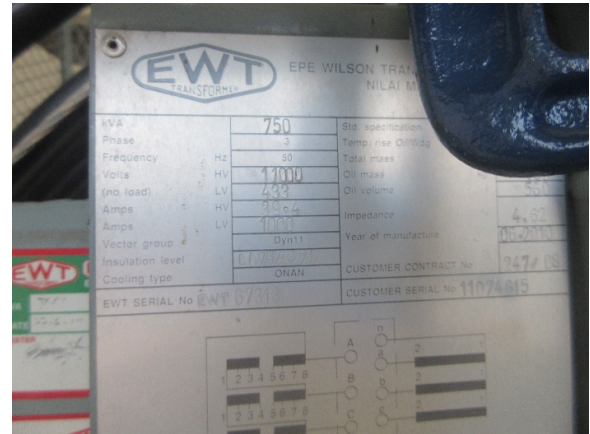


SUBSTATION CONDITION

TRANSFORMER 2



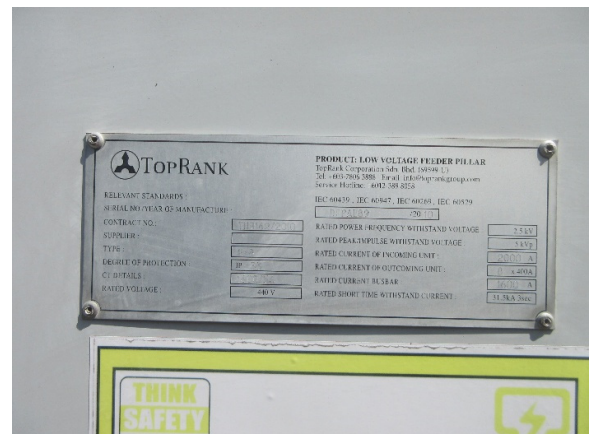
TRANSFORMER 2 NAMEPLATE



FEEDER PILLAR TX1



FEEDER PILLAR TX1 NAMEPLATE



FEEDER PILLAR TX2

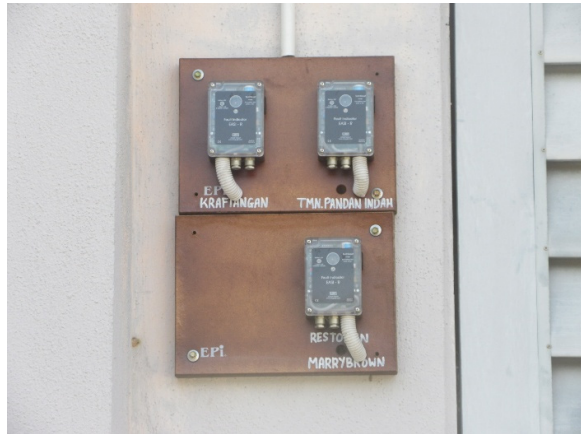


FEEDER PILLAR TX2 NAMEPLATE



SUBSTATION CONDITION

EFI



SF6 INDICATOR



**FIRE EXTINGUISHER
(SWITCHGEAR ROOM)**



FIRE EXTINGUISHER EXPIRY DATE



**FIRE EXTINGUISHER
(TX1 ROOM)**



FIRE EXTINGUISHER EXPIRY DATE



SUBSTATION CONDITION

**FIRE EXTINGUISHER
(TX2 ROOM)**



FIRE EXTINGUISHER EXPIRY DATE



TRANSFORMER 1 OIL LEVEL INDICATOR



TRANSFORMER 2 OIL LEVEL INDICATOR



VISUAL DEFECT

SUBSTATION



CPR POSTER OLD VERSION – SWG ROOM

FP/LVDB



NO FEEDER NO./FEEDER NAME – FP TX1 - NOT FOLLOW NEW STANDARD

FP/LVDB



NO FEEDER NO./FEEDER NAME – FP TX2 - NOT FOLLOW NEW STANDARD

NORMAL/DEFECT STICKER

NORMAL STICKER

CONDITION ASSESSMENT	
	DIAGNOSTIC UNIT SUBSTATION 
Cycle/Year :	1 / 2026
Testing Date :	03-03-2026
Defect :	-
Company/Team :	EET ENGINEERING
	TEV UVS IR

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1

DISTRIBUTION SUBSTATION (PCE) CMA TEST SHEET

STATE / AREA: PAHANG / KUANTAN

PE NAME: K23 RAYA BESAR

DATE: 3/3/2026

TIME IN / OUT: 11:03 A.M. / 11:28 A.M.

FUNCTIONAL LOCATION: CKIN/PCE/1701080

V.I. REF. NO:

TEV BACK GROUND / HUMIDITY: 2 / 1 / 70

BACKGROUND TEMP: 33.1

DETAIL										TEMPERATURE					ULTRASOUND			TEV		REMARK				
NO.	PDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE / SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLIV)	COMP	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE / CYCLE		TYPE CHAR.			
1	CKN 03071	KRAFTANGAN	TRIP / CLOSE		XLPE PILC N/A			CABLE	32.6	32.6	0.0	32.6	176		-6	-	-	-	4	0	-			
								BREAKER																
								TOP PANEL																
								PT																
2	CKN 03074	TMN. PANDAN WDAH	TRIP / CLOSE		XLPE PILC N/A			CABLE	32.5	32.7	0.2	32.6	177		-6	-	-	-	4	0	-			
								BREAKER																
								TOP PANEL																
								PT																
3	CKN 03779	RESTORAN MARRY BROWN	TRIP / CLOSE		XLPE PILC N/A			CABLE	32.6	32.9	0.3	32.8	178		-6	-	-	-	4	0	-			
								BREAKER																
								TOP PANEL																
								PT																
4	CKN 03072	TX1 1000 KVA	TRIP / CLOSE		XLPE PILC N/A			CABLE	32.6	33.0	0.4	32.9	179		-6	-	-	-	3	0	-			
								BREAKER																
								TOP PANEL																
								PT																
OVERVIEW										LOCAL TRANSFORMER														

DETAILS										THERMOGRAPHY					ULTRASOUND		DETAILS										THERMOGRAPHY					ULTRASOUND		REMARKS
TX ID	SERIAL NO	CABLE TYPE / SIZE	REMARK	COMP.	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE / SIZE	COMP.	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS										
TX1	718 834	XLPE/PILC		HT CABLE	37.4	37.8	0.4	37.7	192	-6	-	TX3		XLPE/PILC	HT CABLE																			
		XLPE/PILC	HT BUSHING	41.9	42.2	0.3	42.0	191	-6	-	XLPE/PILC			HT BUSHING																				
		XLPE/PILC	LV CABLE	34.0	34.2	0.2	34.1	190	-6	-	XLPE/PILC			LV CABLE																				
		XLPE/PILC	LV BUSHING	37.0	37.8	0.8	37.4	189	-6	-	XLPE/PILC			LV BUSHING																				
		XLPE/PILC	BODY	36.7	44.5	7.8	40.6	188	-6	-	XLPE/PILC			BODY																				
TX2	EWT 67313	XLPE/PILC		HT CABLE	39.6	39.8	0.2	39.7	199	-6	-	TX4		XLPE/PILC	HT CABLE																			
		XLPE/PILC	HT BUSHING	42.0	42.3	0.3	42.2	198	-6	-	XLPE/PILC			HT BUSHING																				
		XLPE/PILC	LV CABLE	33.4	33.8	0.4	33.6	197	-6	-	XLPE/PILC			LV CABLE																				
		XLPE/PILC	LV BUSHING	35.0	35.7	0.7	35.3	196	-6	-	XLPE/PILC			LV BUSHING																				
		XLPE/PILC	BODY	37.6	45.0	7.4	41.3	195	-6	-	XLPE/PILC			BODY																				

LV DISTRIBUTION BOARD / FEEDER PILLAR / OTHERS

1	LYDB / FP	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
1	LYDB / FP	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC
2	LYDB / FP	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC

NO	LYDB / FP	DEFECTIVE FEEDER NAME	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	REMARK
	INC 1							
	INC 2							
	F1							
	F2							
	F3							
	F4							
	F5							
	F6							
	F7							
	F8							

Battery Bank

Black Box 1

Black Box 2

Lightning Arrestor

Dropout Fuse

Auto Recloser

LBS

EET ENGINEERING SERVICES SDN BHD

No. 38-2, 2nd Floor, Jalan SS 19/1D

47500 Subang Jaya, Selangor.

Tel: 603-5636 6205 Fax: 603-5636 7205

Tested by (Name & Signature):

TNB Supervisor/Engineer (Name & Signature):

MOHD AZZAD BIN GHAMLAN (10658230)

JURUTEKNIK TINGGI AT 1645A

(DIAGNOSTIC SUBSTATION-PAHANG)

SBU MAINTENANCE EAST

DISTRIBUTION NETWORK DIVISION

Ver. : 1-2019

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STATE / AREA:		PAHANG / KUANTAN		DISTRIBUTION SUBSTATION (PCE) CBA TEST SHEET										DATE		5/3/2026		FUNCTIONAL LOCATION:		CKW/PCE/J01080								
FE NAME:		KK3 PATA BESAR		TIME IN / OUT										11:03 A.M.		11:28 A.M.		BACKGROUND TEMP:		33.1 °C								
V.I. REF. NO				TEV BACK GROUND / HUMIDITY																								
DETAIL				SWITCHGEAR										TEMPERATURE				ULTRASOUND				TEV		REMARK				
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLIV)	COMP	Tmin (°C)	Tmax (°C)	ΔT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE/CYCLE	TYPE CHAR.	REMARK							
1	CKN 03073	TX2 750 KVA	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER	32.1	32.5	0.4	32.3	181		-6	-	-	2	0	-								
2			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER																				
3			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER																				
4			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER																				
OVERVIEW																												
LOCAL TRANSFORMER																												
DETAILS					THERMOGRAPHY					ULTRASOUND		DETAILS					THERMOGRAPHY					ULTRASOUND		REMARKS				
TX ID	SERIAL NO	CABLE TYPE/ SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	ΔT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/ SIZE	COMP.	Tmin(°C)	Tmax(°C)	ΔT	FILE	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS				
TX1		XLPE/PILC		HT CABLE								TX3		XLPE/PILC	HT CABLE													
		XLPE/PILC		HT BUSHING										XLPE/PILC	HT BUSHING													
		XLPE/PILC		LV CABLE										XLPE/PILC	LV CABLE													
		XLPE/PILC		LV BUSHING										XLPE/PILC	LV BUSHING													
TX2		XLPE/PILC		HT CABLE								TX4		XLPE/PILC	HT CABLE													
		XLPE/PILC		HT BUSHING										XLPE/PILC	HT BUSHING													
		XLPE/PILC		LV CABLE										XLPE/PILC	LV CABLE													
		XLPE/PILC		LV BUSHING										XLPE/PILC	LV BUSHING													
				BODY											BODY													
LV DISTRIBUTION BOARD/FEEDER PELLAR/OTHERS																												
1	LVDB / FP 1	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10														
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC														
2	LVDB / FP 2	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10														
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC														
NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	ΔT	FILE	Avg (°C)	REMARK							LVDB / FP 1							J-SLOTTED / DIN TYPE						
	INC 1														Manufacturer:													
	INC 2														Serial No.:													
	F1														Rating:							1600A / 1000A / 800A						
	F2														LVDB / FP 2							J-SLOTTED / DIN TYPE						
	F3														Manufacturer:													
	F4														Serial No.:													
	F5														Rating:							1600A / 1000A / 800A						
	F6																											
	F7																											
	F8																											
Battery Bank Black Box 1 Black Box 2 Lightning Arrester Dropout Fuse Auto Recloser LBS																												

EET ENGINEERING SERVICES SDN BHD
 (534556-W)
 No 38-2 / 2nd Floor, Jalan SS 19/1D
 47500 Subang Jaya, Selangor.
 Tel: 603-5636 6205 Fax: 603-5636 7205

MOHD AZZAD BIN OTMAN (10958230)
 JURUTEKNIK TINGKAT II BIASA
 (DIAGNOSTIC SUBSTATION-PAHANG)
 SBU MAINTENANCE EAST
 DISTRIBUTION NETWORK DIVISION

Tested by (Name & Signature):

THB Supervisor/Engineer (Name & Signature):

160

(PP(0-V, 3WGR) , FP(N. label)
TX1, TX2

VISUAL INSPECTION CHECKLIST FOR 11KV SUBSTATION

DATE:

P/E Name (ERMS):

P/E Name (SITE):

GPS Coordinate:

PCE Type:

PCE Area

Switchgear Type 1:

Manufacturer/Modal:

Switchgear Manufactured Date:

Switchgear Type 2:

Manufacturer/Modal:

Switchgear Manufactured Date:

No. of Transformer:

Tx 1:

Tx 2:

Tx 3:

Tx 4:

KEJANGGALAN

1. Signboard:

2. Earth connection:

3. EFI Functioning:

4. Heater Functioning:

5. Labelling @ Switchgear:

6. Labelling @ LVDB:

7. Gas Pressure:

8. Tx Guard:

9. Tx Oil Level (conservator tank):

10. Tx Oil Leak:

11. Tx Bushing:

12. Top Demand Indicator (@LVDB):

13. Link/fuse Holder:

14. LV Termination:

15. Rubber Mat:

16. Trenching:

17. Battery Voltage:

18. Lighting:

19. Roof Condition:

20. Fire Fighting

21. Fence/door condition:

22. Cleanliness/Bushes:

23. Vandalisme

24. Other Findings

3 / 3 / 2026			
KK3 PAYA BESAR			
KK3 PAYA BESAR			
3-788662, 103-239089			
INDOOR	☒	OUTDOOR	☐
NORMAL	☒	INDUSTRIAL	☐
RMU OIL	☐	RMU SF6	☒
WPKOM / W524			
2014			
RMU OIL	☐	RMU SF6	☐
2			
Type	HS	KVA	1000
Type	HS	KVA	750
Type		KVA	
Type		KVA	

F/L NO (ERMS):

F/L NO (SITE):

CKTN / PCE / J01080

CKTN / PCE / J01080

ATTACH BUILDING

PAT

COASTAL

MRMU

VCB

OCB

Switchgear Rating:

12 kV

RMU SF6 / OIL S/N:

14093213346DE

Switchgear Rating:

RMU SF6 / OIL S/N:

Construction Yr:

2010

Manufacturer:

SGB

S/N:

718834

Construction Yr:

2010

Manufacturer:

EWT

S/N:

EWT 67313

Construction Yr:

Manufacturer:

S/N:

Construction Yr:

Manufacturer:

S/N:

YES	☒	NO	☐	Remarks
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
Good	☐	Not Good	☒	Remarks N/A
Good	☒	Not Good	☐	Remarks
Good	☐	Not Good	☒	Remarks NO label
Good	☒	Not Good	☐	Remarks
YES	☒	NO	☐	Remarks
Good	☐	Not Good	☒	Remarks N/A
YES	☐	NO	☒	Remarks
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
YES	☒	NO	☐	Remarks
Good	☒	Not Good	☐	Remarks
Good	☐	Not Good	☒	Remarks N/A
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
Valid	☒	Expired	☐	Remarks Expiry Date: 22/9/2026 No. of missing portable: -
Good	☒	Not Good	☐	Remarks
Good	☒	Not Good	☐	Remarks
YES	☐	NO	☒	Remarks

EET ENGINEERING SERVICES SDN BHD

(534556-W)

No. 38-2, 2nd Floor, Jalan SS 19/1D

47500 Sepang, Selangor.

Tel: 603-5656 6205 Fax: 603-5656 7205

TESTED BY (NAME & SIGNATURE):

TNB SUPERVISOR/ ENGINEER (NAME & SIGNATURE):

MOHD AZZAD BIN OTHMAN (10658230)
JURUTEKNIK TINGGI AT BIASA
(DIAGNOSTIC SUBSTATION-PAHANG)
SBU MAINTENANCE EAST
DISTRIBUTION NETWORK DIVISION